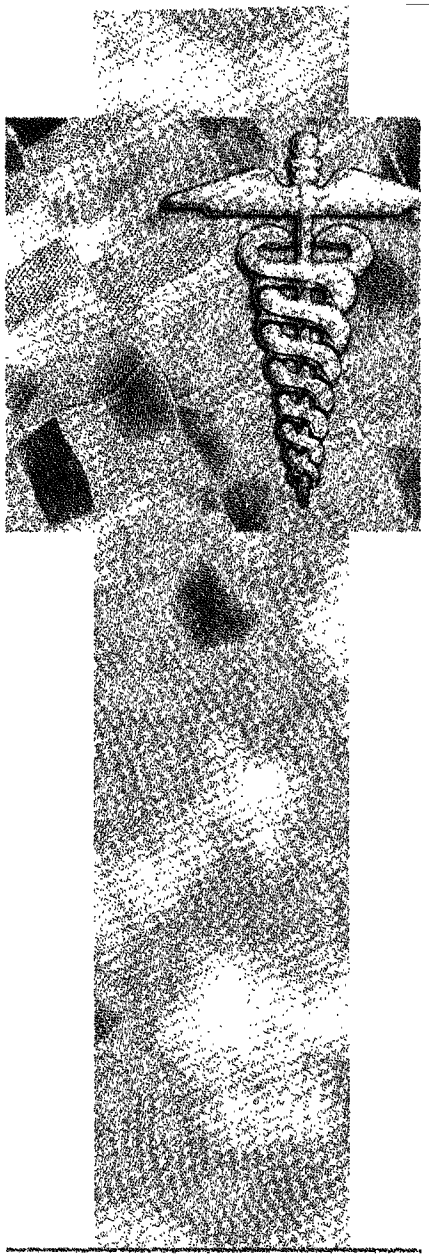


Developments in ECG Acquisition, Preprocessing, Parameter Measurement, and Recording

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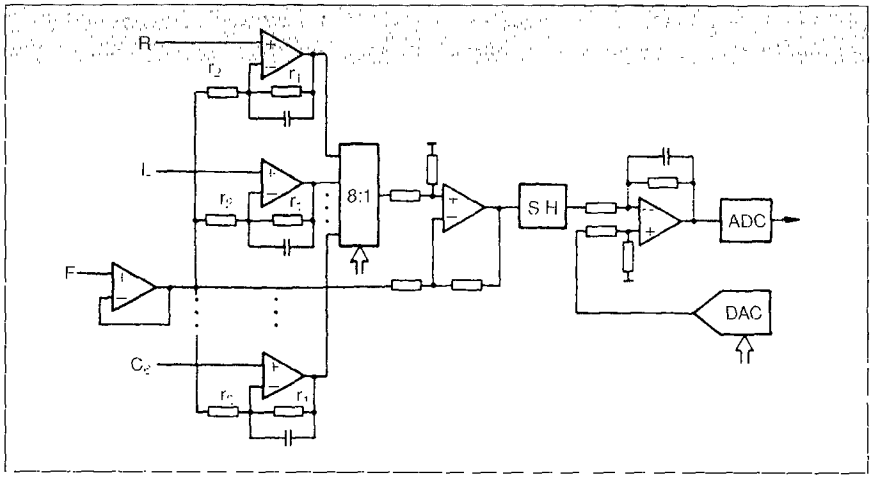
Our development of microprocessor-based electrocardiographs started at the beginning of the 1980s. The first publication dealing with a complete instrument of this type was presented by us in 1985 [1]. Its implementation resulted from the decision to use a microprocessor-controlled microdot thermal printer instead of the standard at that time, which was the heated stylus type recorder with its inherent distortions and need of frequent service.

The next logical step was to perform as much of the instrument functions as possible under software control. Simultaneous acquisition of the 12 standard leads was implemented and combined with signal preprocessing, wave detection, and electrocardiogram (ECG) parameter measurement. Further developments [2] involved separate microprocessor control modules for signal acquisition, processing, and printing/plotting. Modules connected to personal computers were also used for design of sophisticated ECG systems.

The subjects considered in this article follow the logical procedures of acquisition, preprocessing, wave detection, parameter measurement, and recording of the ECG. We present an economic procedure for simultaneous multichannel signal acquisition, digital real-time interference elimination and drift suppression, a real-time software QRS detector, and algorithms for wave onset and termination determination. We also discuss fast algorithms for lengthwise and crosswise lead presentation that supports the use microdot thermal printers for signal recordings.

Signal Acquisition

The instrumentation amplifier is commonly used for ECG acquisition. Multichannel instruments require a large number of operational amplifiers. This might cause difficulties, especially with floating input circuitry. An economic multichannel structure has been proposed [3] with only one noninverting preamplifier per channel. The driven right-leg configuration was used for interference compensation, but the body-driving amplifier must meet special requirements.



1. An 8-channel DC amplifier (from which the 12 leads are derived), featuring just 10 operational amplifiers.

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Another version (Fig. 1) using a small number of operational amplifiers was developed [4, 5]. This structure ensures a high common mode rejection ratio (CMRR) for each pair of differential input signals. The left leg (F) potential is buffered and applied to the common point. The amplified voltages at the first stage outputs are the potentials of the right and left arms (R and L) and those of the chest positions (C_1 to C_6) with respect to **F**:

$$U_R = R + \frac{r_1}{r_2}(R - F)$$

$$U_L = L + \frac{r_1}{r_2}(L - F)$$

...

$$U_{C_6} = C_6 + \frac{r_1}{r_2}(C_6 - F)$$

These voltages are then multiplexed and the F potential subtracted, thus obtaining a transition to asymmetrical output. Eight differences, U_{R-F}, U_{L-F}, $U_{C1-F},...,U_{C6-F}$, are obtained. Finally, the 12 standard leads are easily computed (avoiding the use of resistors for acquiring Goldberger and Wilson reference potentials), as shown below:

$$I = U_{L-F} - U_{R-F} = (L - R)\left(1 + \frac{r_1}{r_2}\right)$$

...

$$aVR = U_{R-F} - \frac{U_{L-F}}{2}$$

...

$$V_6 = U_{C_6-F} - \frac{U_{R-F} + U_{L-F}}{3}$$

We call the differences, U_{R-F}, U_{L-F}, $U_{C1-F},...,U_{C6-F}$, the primary leads, which are further referred to as **R, L, C_1,...,C6**.

Inadequate high-pass filtering has been known to introduce distortions in the ST segment [6, 7]. Therefore, a first-order high-pass filter with 3.2 sec time constant has been universally accepted in order to preserve ECG waveforms.

In our opinion, the main problem in high-pass filtering is that the QRS complex induces distortions, especially when it is monopolar, of high amplitude, and long duration. We have shown [8] that partial differentiation of such QRS complexes can occur, resulting in an after-effect component superimposed on the initial part of the ST segment and changing its shape. In many medical ECG interpretation textbooks [9-11], ST shifts opposite to high monopolar QRS complexes are wrongly described as additional signs of hypertrophy or of some types of blocks.

Recently, we studied in more detail the low-frequency distortions caused by the standard 3.2 sec time constant high-pass filter [12]. These distortions can be compensated by digital backward filtering, but the procedure is not applicable in real time. Also taking into account the long settling time after artifacts for AC amplifiers, we decided that DC amplification is the best solution.

The DC amplifier of Fig. 1 was first used for an electroencephalograph [5]. The digital-to-analog converter (DAC) output is software controlled and used for subtraction of the electrode-skin offsets from the input signals. A specific procedure compensates for the amplified DAC error when the code changes during signal acquisition.

The DC amplifier is an ideal instrument for fast recovery and accurate recording of the ECG after defibrillation, pacemaker pulses, or other artifacts. The result of a defibrillation test, according to the recommendation IEC 62D(CO)6A [13, 14], is shown in Fig. 2. It can be seen that in less than 1 sec adequate signal recording is resumed. In pacemaker patients, high fidelity recordings are obtained (Figs. 3, 4), where spontaneous, paced, and fusion beats can be seen. An additional advantage of the DC amplifier is the possibility for easy recognition of broken leads by detection of resulting channel saturation.

Preprocessing

Power-line Interference Elimination

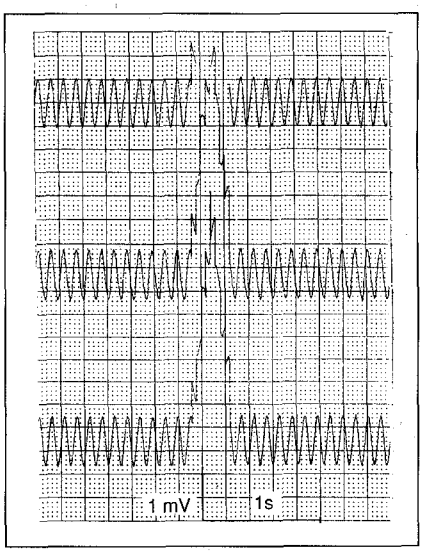
Despite the high CMRR of the ECG input amplifiers, recordings are often contaminated by power-supply interference, usually phase shifted with respect to the mains voltage and containing harmonics. This noise is due to parasite currents through the patient's body and the connecting cables and differences in the electrode impedances, converting common mode interference voltages into unwanted differential signals [15,16]. Therefore, the filtering of power-line interference from the ECG signals is a most important step, preceding all other processing procedures.

The use of conventional filters affects the signal components adjacent to the mains frequency. A variety of adaptive filters to prevent this effect were proposed [17-21], but they require an adaptation period after each interference or abrupt signal change, and they are not as effective in cases of extra systoles and arrhythmias.

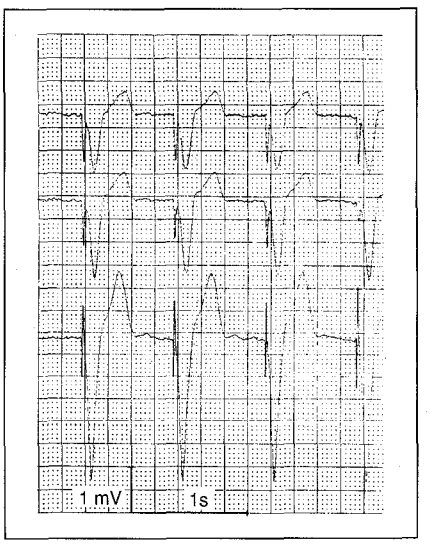
These inconveniences are effectively avoided by the use of a digital subtraction procedure [22, 23], essentially consisting of the following steps:

1. The ongoing period of the mains voltage, T_M, is measured and the sampling period, $t_S = T_M/n$ is defined; where $n = f_S/f_M$ is the number of samples in T_M, and f_S and f_M are the sampling rate and the rated mains frequency, respectively. For 50 Hz, $n = f_S/50$.

2. Short segments (usually less than 40 ms), in which the ECG signal evolves linearly, are detected using a criterion based on

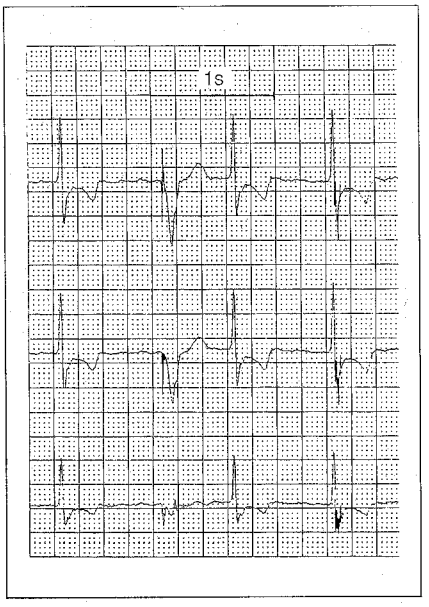


2. DC amplifier signal recovery test after defibrillation (IEC recommendation [14]).



3. DC amplifier recording of paced ECG.

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4. DC amplifier recording of paced, spontaneous, and fusion beats.

differences between samples taken at intervals equal to or multiples of $\mathbf{T_M}$.

3. Moving averaging over **n** samples is applied. The discrete values thus obtained are samples without interference, due to the fact that the first zero of this averaging filter is exactly at $\mathbf{f_M}$.

4. The samples without interference are used to compute **n** amplitudes of the interference, phase locked to $\mathbf{T_M}$, called phase corrections.

5. The corrections are subtracted from the corresponding samples of the signal in nonlinear segments. Thereby, the original signal spectrum is preserved.

As the next zeroes of the moving averaging filter are multiples of $\mathbf{f_M}$, the procedure effectively deals with mains interference of arbitrary (nonsinusoidal) waveforms, as is the case in practice. A selected ECG signal of varying shape (Fig. 5) is used to point out that the procedure does not introduce any waveform distortions. The upper trace is the original signal, the middle shows the same signal after applying the procedure, and the lower trace is the difference between the two.

The mains interference may change in amplitude and frequency during recording. Due to the corrections update in each linear segment of the ECG signal, the procedure is adaptive and can cope with amplitude fluctuations. The efficiency is closely related to the criterion used. Less strict criteria (including some linear portions in the P and T waves) permit the detection of more linear segments in a cardiac cycle, leading to more frequent updating of the correction values and applying the corrections to intervals nearest to the ones where they were computed.

An ECG signal, corrupted by interference, with abruptly changing, constant, and gradually changing amplitudes is presented in Fig. 6 (the traces are arranged as in Fig. 5). It can be seen that the procedure is unable to eliminate abrupt interference transitions only within less than two periods of $\mathbf{T_M}$ (Fig. 6: 40 ms for $\mathbf{f_M}$ = 50 Hz), but such changes are practically impossible in a power system.

The mains frequency fluctuations could also induce errors, which were analyzed and confined to allowable limits [24].

Baseline Fluctuation Suppression

Baseline fluctuations may be considered a less disturbing event, compared to mains interference, but can be more difficult to eliminate or even suppress. The standard first-order high-pass filter of 3.2 sec time constant cannot block most low-frequency baseline drift. It also may bring problems in visual or automatic ECG measurement and interpretation.

Linear-phase two-pass digital filtering, applied consecutively forward and backward on the sampled signal data can substantially reduce distortion. The filter equation is:

$$\mathbf{Y_n = k_1(X_n + X_{n-1}) + k_2 Y_{n-1}}$$

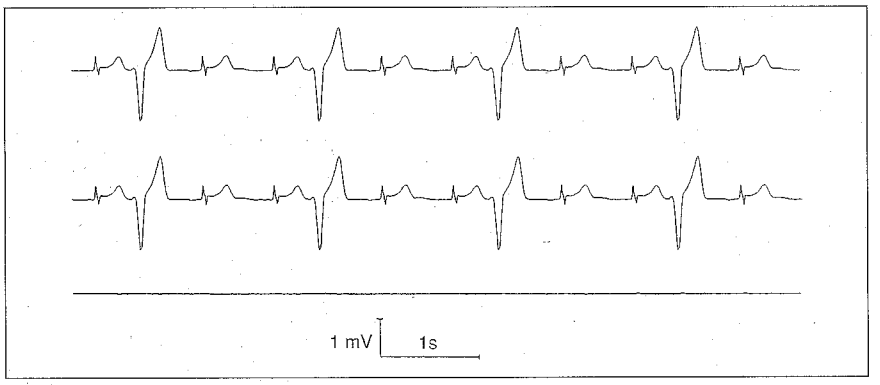
where **X** are nonfiltered samples, **Y** are filtered samples, and:

$$\mathbf{k_1 = \frac{1}{1+tg(\pi fT)} \quad k_2 = \frac{1-tg(\pi fT)}{1+tg(\pi fT)}}$$

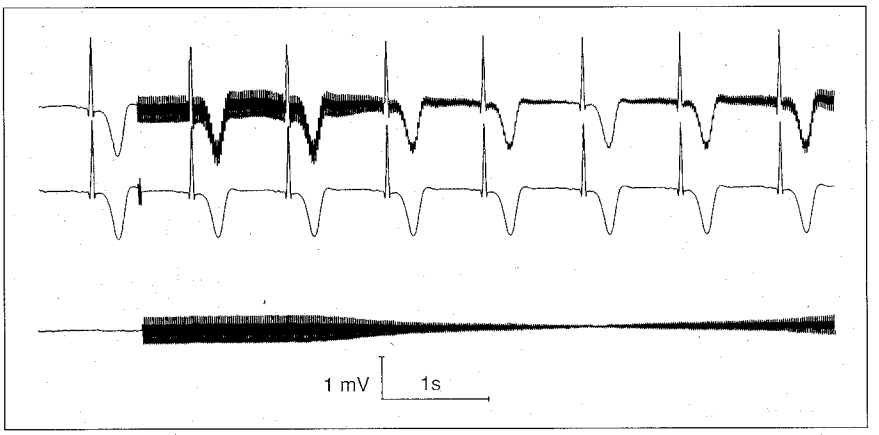
where **f** is the cut-off frequency in radians and **T** is the sampling period.

We implemented this filter with **f**=0.64 Hz cut-off and obtained very good results [2]. The drawback of this method is that it cannot be applied in real time.

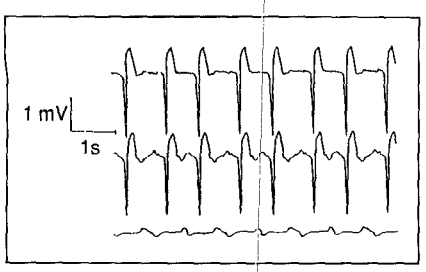
Later, we designed a procedure [25] that applies fast linear-phase low-pass filtering to extract the drift and subtract it from the original signal. This process can be done with a short and constant delay, allowing a quasi real-time mode of operation. However, without the use of addi-



5. An original ECG signal (upper trace) subjected to the power-line interference subtraction procedure (middle trace). No distortions are introduced, as shown by the difference of the two signals (lower trace).



6. Abrupt switched-on and changing interference on the original signal (upper trace). Middle trace: processed signal. Lower trace: Removed interference.



7. A high and wide QRS signal (upper trace) subjected to 0.4 Hz cut-off low-pass filtering (middle trace) and the difference signal (lower trace), showing residual waves.

tional means, higher slew-rate waves (normally the QRS complexes) increase errors, as can be seen in Fig. 7. Here, the averaging filter has a first zero at 0.8 Hz, corresponding to $f \approx 0.4$ Hz cut-off. The distortions are evident in the lower trace, representing the difference between nonfiltered (upper trace) and filtered (middle trace) signals. Therefore, we devised a procedure for removal of the QRS complexes (all waves with slope >2.5 μV/ms) prior to filtering. These eliminated signal portions are interpolated by linear segments.

The drift reduction process was assessed by adding a 0.5 Hz 0.7 mV sinusoidal "drift" to the original signals. This relatively high frequency can be considered a difficult experimental condition. The result is shown in Fig. 8. The higher slew rate waves detected are marked by short upward and downward directed slashes. Note that virtually no distortions correlated with the heart rate appear in the processed signal. However, it is not purely sinusoidal. This is due to some longer high slew-rate intervals that are eliminated by linear interpolation. Sometimes, drift added to a wave may be detected as a high slew-rate signal. Therefore, in a later version, the procedure was improved by applying the slope criterion within the QRS width only, using a real-time QRS detector.

Wave Detection and Parameter Measurement

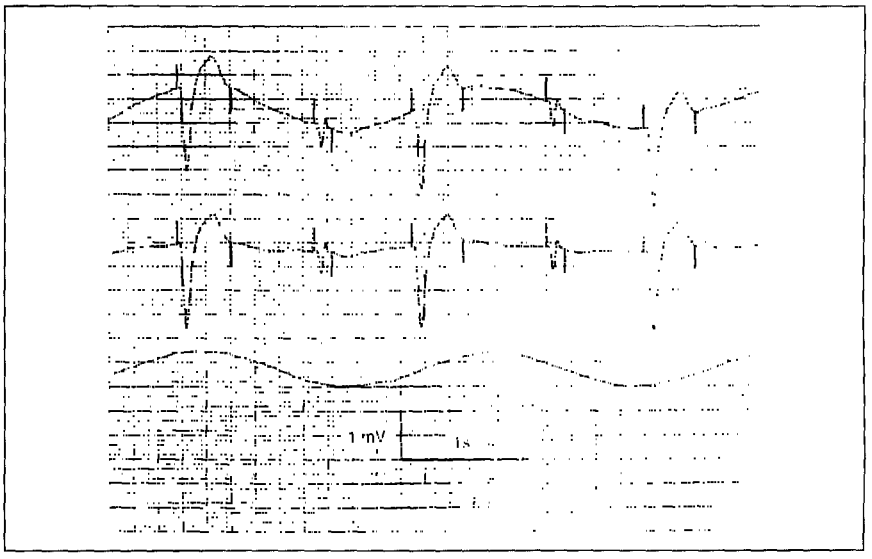
QRS Detection

Automated ECG analysis requires wave amplitude and duration measurements. Therefore, wave edge detection is mandatory, with accurate location of onset and termination points. This process begins with identification of the QRS complex—the most prominent wave in the R-R interval.

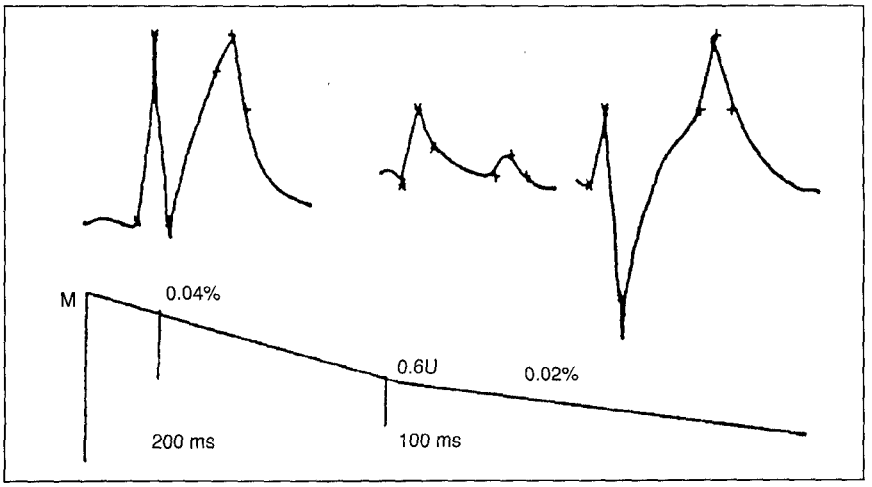
QRS detector procedures consist of two main phases: preprocessing and application of decision rules [26]. The QRS complexes are enhanced by filtering and nonlinear transformation, while interference and other ECG signal components are suppressed. Decision rules classify some of the waves as QRS complexes. This procedure can be optimized by trial-and-error tests in order to obtain an increased number of correct detections. The preprocessing is normally aimed at achievement of a high signal-to-noise ratio.

Friesen, et al. [27], presented a comparison of nine QRS detection algorithms, based on: i) amplitude and first derivative, ii) first derivative only, iii) first and second derivative, and iv) digital filtering. We applied these algorithms to selected signals from our data base containing records with pronounced baseline drift. The results were unsatisfactory. This failure is probably due to the use of fixed detection thresholds, whereas adaptive ones would be more appropriate.

The diversity of algorithms for QRS detection and the continuous efforts for improvement show unambiguously that it is highly unlikely to find one solution covering all waveforms. We designed a fast and simple algorithm for QRS detection in real time [28]. Our previous investigations have shown that the QRS complex can be selected from among other candidates by means of the following parameters: i) high amplitude, ii) steep leading and/or trailing slopes, and iii) sharp peaks.



8. An original signal with an added sinusoid of 0.7 mV 0.5 Hz (upper trace), the processed signal (middle trace), and the removed drift (lower trace).



9. Three specially selected QRS-T complexes of unusual shape are shown. Detected intervals containing sharp peaks are marked by an asterisk (upper panel). The time course of the changing threshold, M, is presented in the lower panel.

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A combined criterion was developed. Two differences between the current sample, S_i, and the samples S_{i-n} and S_{i+n} are calculated. If both are of the same sign, the unsigned sum, U_i, is the current nonzero value of the combined criterion marking a detected peak.

$$U_i = \begin{cases} |2S_i - S_{i-n} - S_{i+n}| \\ \text{for sign } (S_i - S_{i-n}) = \text{sign } (S_i - S_{i+n}) \\ 0 \\ \text{for sign } (S_i - S_{i-n}) \neq \text{sign } (S_i - S_{i+n}) \end{cases}$$

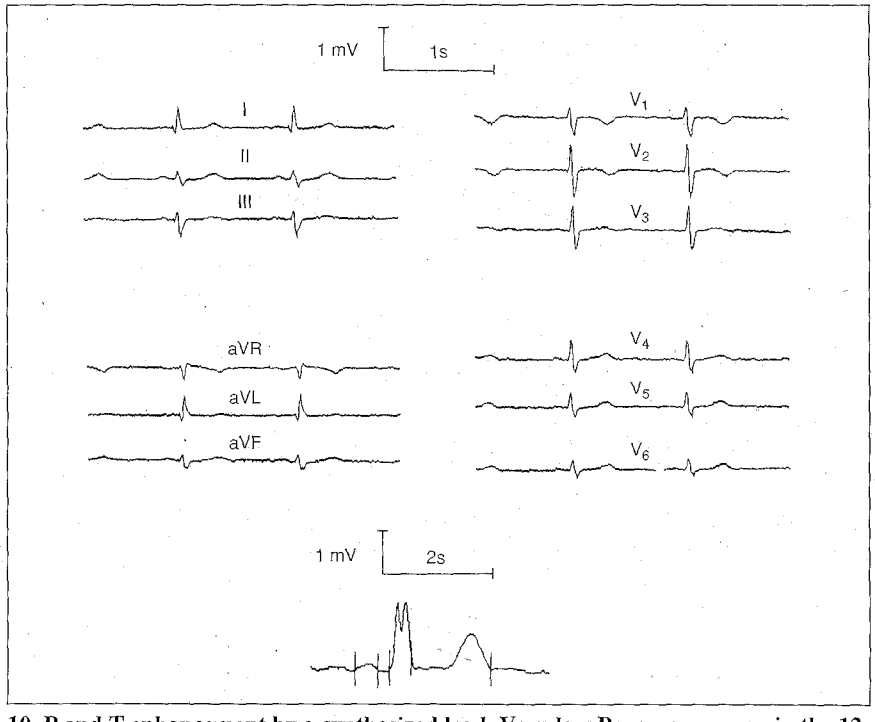
Then, U_i is compared with an adaptive threshold, **M**, and is considered to be a QRS complex if $U_i > M$.

The intersample distance, **n**, should be appropriately selected. Our experience suggests a time interval of 20 ms to have the additional advantage of avoiding mains interference.

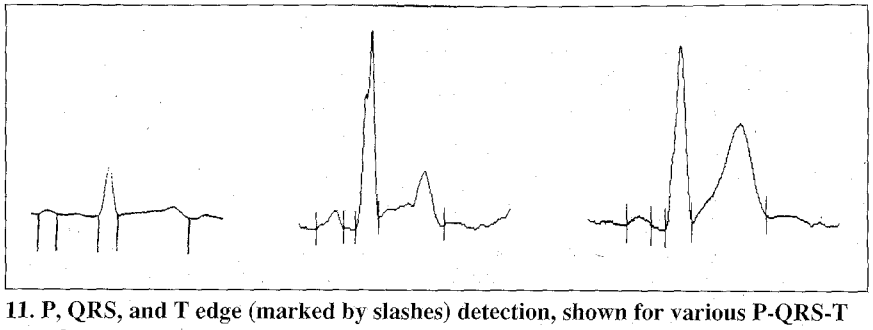
Figure 9 (upper panel) shows three QRS-T intervals, selected as difficult for identification because of their unusual shapes. The superimposed marks prove, that the **U** criterion for QRS complexes (*) is greater compared to that of, e.g., high T waves (+). The lower panel illustrates the time-dependent threshold course, **M**. After the first second, the decrease of **M** is two times slower. According to the generally accepted rules, a dead time of 200 ms is introduced.

A ventricular beat detector and classifier [29] was also developed and tested. The procedure includes computing the pseudospatial velocity, as obtained from II, V_2, and V_6 leads. Then, an iterative decision rule is applied. Finally, the fractal dimension is computed, allowing for accurate off-line detection and classification of various ectopic beat waveforms.

There are cases when accurate position of the detected fiducial point with respect to the QRS is required, or when RR intervals are to be measured with high accuracy. If the higher sampling rate thus required is inconvenient (e.g. portable ECG recorders with limited memory; ECG databases of low sampling rate—MIT-BIH of 360 Hz, etc.), another solution is possible [30]. The low sampling rate signal is subjected to comb filtering (for interference suppression, first zero at mains frequency) and a cubic approximation is applied. Then the signal is resampled at higher frequency. This approach improves the resolution of RR interval measurement to 1 ms, from signals sampled at 250, 200, or even 100 Hz. It also allows for heart rate variability spectral analysis of higher accuracy.



10. P and T enhancement by a synthesized lead. Very low P waves are seen in the 12 leads, compared with the V_{PT} lead.



11. P, QRS, and T edge (marked by slashes) detection, shown for various P-QRS-T complexes.

Wave Onset and Termination Detection

After adequate preprocessing and successful QRS detection, the first major problem of wave measurement is finding a zero-level reference point, Z_O. A QRS edge-detection procedure is applied, based on assessment of the slope from the R-wave peak (the maximum found in a 100 ms time interval around the QRS detection fiducial point) to the left. A slope criterion, $C_S \leq 10\ \mu V/ms$ for 20 ms, is applied, thus finding a "flat" interval, within which Z_O is obtained as an average value. If no R peak is detected, the S peak is taken.

The next problem is to detect the P wave, the QRS complex onset and termination, and the end of the T wave. Our experience suggests that these points (the P-QRS-T complex time-parameters) should be searched for in a compound lead; for example, the ECG spatial vector magnitude, V_{SP}, or another all-inclusive signal, rather than in the separate leads. As the computation of V_{SP} may be a problem with some microprocessor systems, we tried several simplified synthesized leads and tested their performance.

An approximate procedure for obtaining V_{SP} involves three quasi-orthogonal leads, **X**, **Y**, and **Z**, where
$X = 0.5[(C_4 - Z_O) - (C_1 - Z_O)]$,
$Y = -(R - Z_O)$,
$Z = [(R - Z_O) - (C_2 - Z_O)]$ and R, C_2, C_4

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are the primary lead signals (see the Signal Acquisition section). Then:

$$V_{SP} = 0.58[(|X|+|Y|+|Z|) + 0.18(||X|-|Y||+||X|-|Z||+||Y|-|Z||)]$$

Alternatively, two different compound leads can be used, one for QRS (V_R) and the other for P and T waves enhancement (V_{PT}). V_R is a kind of spatial velocity vector, obtained as a sum of the derivatives of the eight primary lead signals:

$$V_R = \sum_{n=1}^{8} (S_{n,i+1} - S_{n,i})$$

where $S_{n,i}$ is the i-th sample of the lead **n**.

V_{PT} is obtained as the sum of rectified or squared amplitudes of the primary leads, **R**, C_1, and C_2:

$$V_{PT} = |S_{R-F,i}| + |S_{C_1-F,i}| + |S_{C_2-F,i}|$$

$$V_{PT} = (S_{R-F,i})^2 + (S_{C_1-F,i})^2 + (S_{C_2-F,i})^2$$

All amplitudes are measured with respect to the previously found zero reference, Z_O.

The detection accuracy is further improved by a second application of the edge-detection algorithm, this time on V_R or V_{TP} leads, thus finding a more accurate value for Z_O. The QRS onset is taken at the right-most address of the flat interval. The termination is located by the same algorithm, but with a slope criterion of C_S ≤5 μV/ms for 20 ms.

Next, R-R intervals are measured. P and T wave edges are searched for in time-interval windows depending on the R-R interval. The search is done on lead V_{PT} by the same edge-detection algorithm. If no 20 ms flat segments are found, additional low-pass filtering is applied and the procedure repeated.

P and T wave enhancement through V_{PT} is shown in Fig. 10. Here, practically no P waves are detectable in the eight primary leads, but a well-defined wave appears in V_{PT}. The edge points detected are marked on several P-QRS-T complexes of V_{PT} (the QRS edges being taken from V_R) in order to allow assessment of the overall performance of the procedure (Figs. 10, 11).

Some special investigations of the P-wave shape and area or of the vectorcardiogram P-wave loops can be carried out on more enhanced and amplified waves. An example is shown in Fig. 12, where the wave is obtained with a resolution of 0.1 μV/bit and a 1000 Hz sampling rate.

The amplitudes of the P and T waves in the separate leads are easily measured by searching for maxima or minima, using the addresses of the respective edges detected. Amplitudes lower than 30 μV are discarded. A more complicated search is needed for the measurement of waves within a QRS complex, where a variety of 15 waveforms exists, from just R or S waves, through QS and RS types, to rSR´s, QrsR´, and Rsr´s´.

Starting from the R wave (or S wave, if no R is present), a search takes place to the left and right for extrema of the next lower value (with respect to R). From the addresses of these extrema, new subintervals are formed and the same search is applied again, until the entire QRS width is explored. Finally, the extrema found are labeled Q,R,S; Q´,R´,S´; q,r,s; and q´,r´,s´, according to their positions and amplitudes within the complex. Extrema of values below 50 μV are neglected, as accepted in electrocardiogram interpretation.

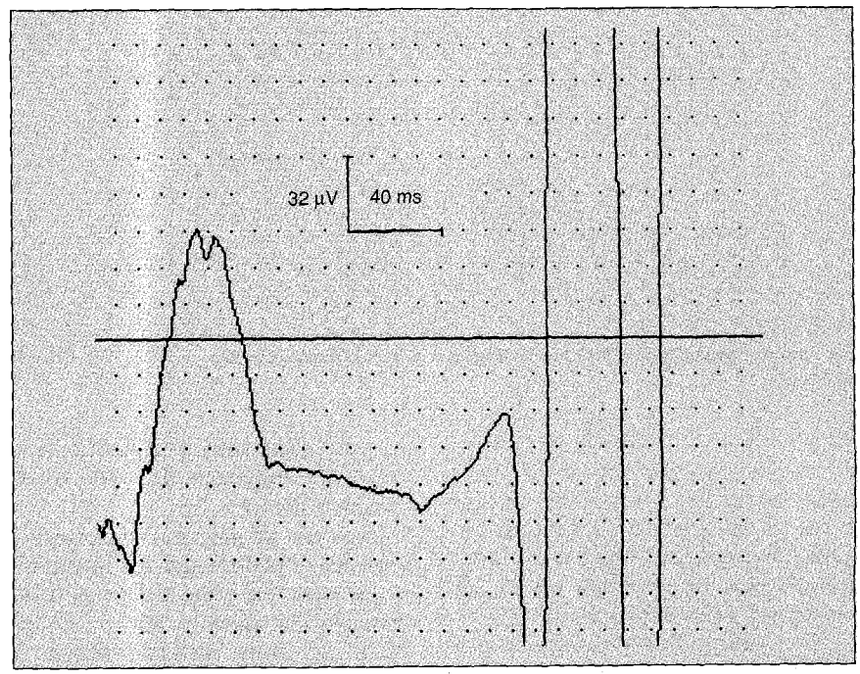
Signal Recording

Electrocardiogram Recording

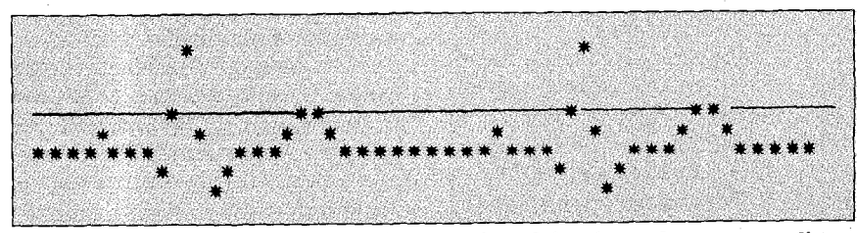
The use of microdot thermal printers (MTPs) is a widespread practice for ECG recording, due to their incontestable advantages [1, 31]. However, there are a few publications dealing with efficient microprocessor MTP control for ECG recording. Therefore, we are tempted to briefly outline our experience.

Regardless of the particular type, every MTP has a thermal head and a paper-driving motor. Due to the large number of dots required (most often 8 per mm), the thermal head includes a shift register, usually loaded from top to bottom by a one-byte parallel interface.

The software problems for controlling an MTP consist of: i) Offset of the sampled channel values to obtain an appropriate trace location on the paper; ii) Setting an additional value related to the sampled value in order to link the next adjacent

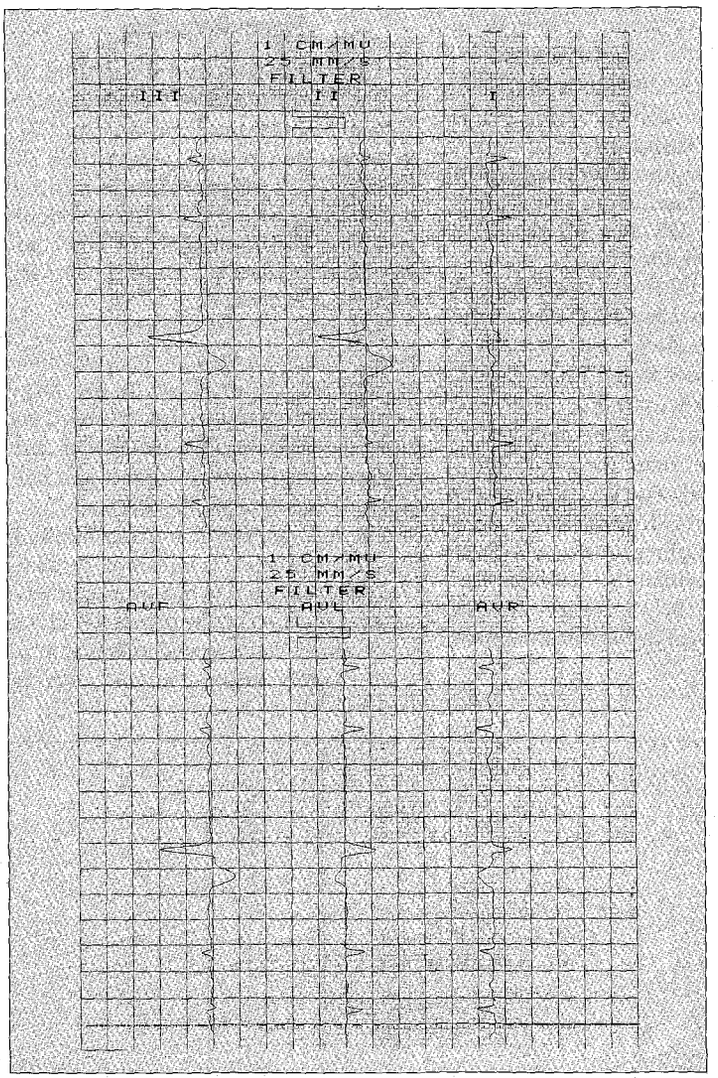


12. Highly amplified P-wave.

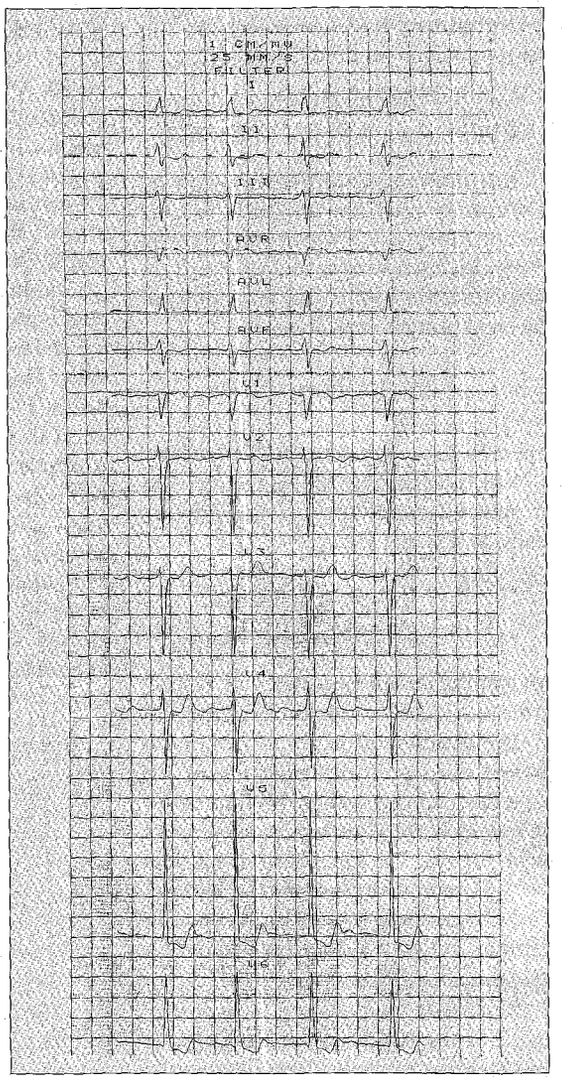


13. Crosswise recording algorithm using detection of signal samples corresponding to successive horizontal lines. One line is shown.

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14. Conventional lengthwise recording (peripheral leads only are shown).



15. Simultaneous 12-channel crosswise recording.

sample whenever it differs by more than 1 dot in amplitude, thus providing smooth traces; iii) Dealing with possible crossing of channel traces; and iv) Graphic line formation by selecting "black dots" in the thermal head to be heated with 1 and nonheated "white dots" with 0.

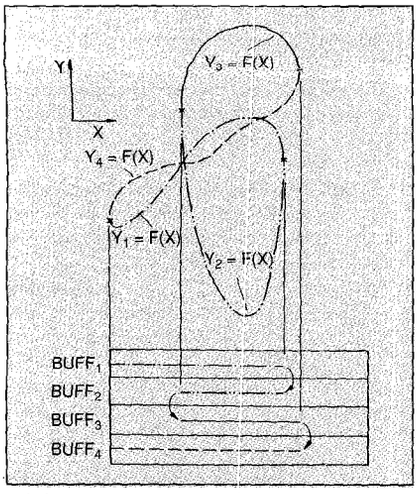
The familiar ECG printing mode is lengthwise to the paper. Nevertheless, recording across the paper has the advantage of more conveniently presenting simultaneously acquired signals in 12 (or even more) channel format.

We developed an algorithm for lengthwise graphic line formation using preliminary computed lengths of "black" and "white" strips. The parallel input of the shift register is loaded with 00000000 or 11111111. The strip value is subtracted by 8 and a data load signal for the shift register is generated every time a borrow did not occur. Otherwise, a "mixed" byte is compiled by shifting the data at the parallel input to the left.

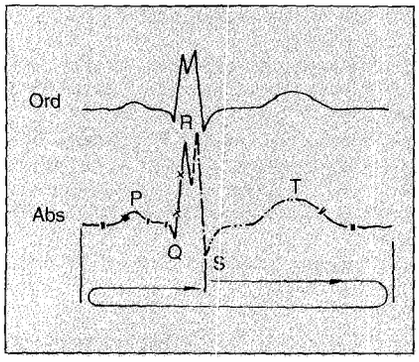
Crosswise printing was implemented by adaptation of the lengthwise algorithm. In this case, the graphic line must be formed by the sample addresses used for "black" and "white" strip computation. A straightforward approach to graphic line formation from a row of samples (e.g., the ones marked by a line in Fig. 13) consists of: i) determination of the row level; ii) comparison of all samples, from left to right, with this level; iii) setting "1" in the locations where coincidence occurs and "0" everywhere else. Unfortunately, the consecutive sample search is a slow process. Creation of linking values (not shown in the figure) could be additionally difficult.

We devised the following very fast algorithm: i) The samples in every row are counted and their sums recorded in memory. ii) The first address of the first row is loaded in the first location of a buffer, **BUF1**. The location with the first address of the second row is then obtained by adding the first row sum, and so on for the next rows. iii) The samples are taken in sequence. Their row numbers show their corresponding location in **BUF1**. This location contains the cell number in a second buffer, **BUF2**, where the address of the current sample is stored. Then, the content of the **BUF1** location is incremented. iv) A "black" strip length is equal to the number of successive sample addresses. A "white" strip length is determined by the difference between two adjacent nonsuccessive sample addresses, decremented by 1. v) An adequate algorithm for linking is added to the above procedure.

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16. Decomposition of a VCG loop into consecutive time intervals, representing non-ambiguous $Y_i = f(X)$ functions.



17. Ordinate (upper trace) and abscissa (lower trace) of the VCG loop.

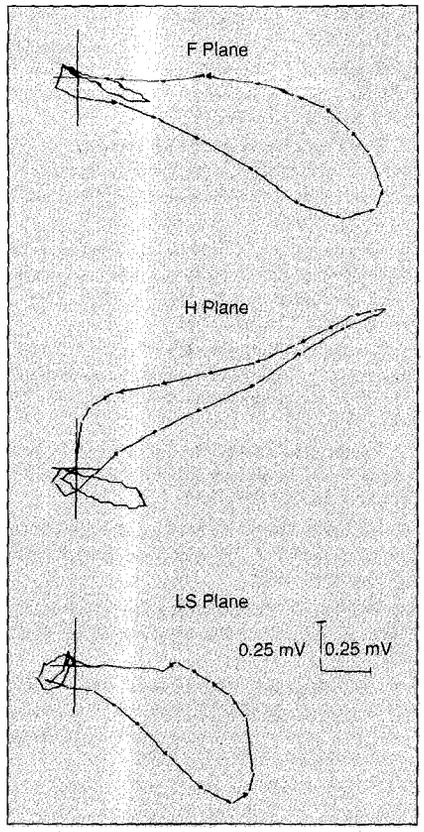
Both recording modes are demonstrated in Figs. 14 and 15. The effective printing width is 100 mm, limiting the lengths of the recorded epochs in crosswise mode to 4 sec for an equivalent paper speed of 25 mm/s and 8 s for 12.5 mm/s.

Vectorcardiogram Loops Recording

Vectorcardiography has been considered in the near past to be a time consuming examination because of the special electrode placement and difficulty in recording. However, orthogonal ECG and vectorcardiogram (VCG) signals can be easily obtained by computation from the standard 12 leads [32]. Our obvious next step was to record VCG loops immediately after the routine ECG examination by MTP [33], using an economical and very fast algorithm.

The basic idea is to decompose the loop into consecutive time intervals, representing nonambiguous $\mathbf{Y_i = f(X)}$ functions (Fig. 16). The intervals are stored in buffers $\mathbf{BUFF_i}$ and considered as $\mathbf{Y_i = f(t)}$ "channel" signals to be recorded. The decomposition is preceded by the following four steps over two orthogonal ECG leads, representing the abscissa (Abs) and the ordinate (Ord) of the loop to be traced (Fig. 17): i) determination of the beginning and the end of a selected P-QRS-T interval; ii) smoothing of the digitized data (except in the QRS domain) in order to suppress possible noise, thus reducing the number of buffers; iii) shifting the abscissa values until the lowest one (usually a Q or S peak) is located in the left most position in the printing area; and iv) shifting the ordinate values until the lowest and highest values correspond to a symmetrical vertical position in the printing area.

The loop tracing direction is marked by arrows on the top of segments, with time duration equal to the intersample interval. A rectangular cross is printed at the isoelectric point. A VCG with its three plane loops, recorded with a gain of 16 cm/mV, is shown in Fig. 18.



18. Vectorcardiogram in the three standard orthogonal planes.

Discussion

The presented developments have been implemented in a family of microprocessor ECGs. Since 1982, more than 6500 units have been manufactured by EMA Engineering-Sofia in different versions. They are used in various clinical settings, including intensive care wards and emergency units, that prove the validity of the scientific investigations and the efficiency of their hardware and software implementation in acquisition, preprocessing, parameter measurement, and recording of the electrocardiogram. The powerful interference elimination and drift suppression procedures developed considerably reduce requirements for high CMRR, shielded cables, careful skin preparation, special electrode gels, etc.

There are trends emerging in electrocardiology and electrocardiography connected with instrumentation and signal analysis. We might speculate that, with the availability of high resolution analog-to-digital converters (16 to 24 bit) and low polarization voltage electrodes, DC amplifiers will soon be favored. In our case (Fig. 1) the need for a digital-to-analog converter and associated software control will become obsolete.

Acknowledgments

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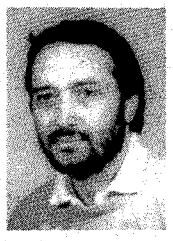
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